Dynamic Pricing Engine Using Machine Learning and Reinforcement Learning

1. Abstract

Dynamic pricing is a strategy used by modern businesses to adjust prices in real time based on demand, competition, and other external factors. This project presents a dynamic pricing engine that combines supervised machine learning models such as Random Forest and Boost with reinforcement learning (Q-learning) to optimize pricing decisions. The system predicts optimal prices and continuously improves pricing strategies using feedback.

2. Introduction

Pricing plays a crucial role in maximizing revenue and maintaining competitiveness. Traditional static pricing models fail to adapt to changing market conditions.

This project aims to:

- Predict optimal prices using ML models
- Learn dynamic strategies using reinforcement learning
- Simulate real-world pricing scenarios

3. Literature Review

Dynamic pricing has been widely used in:

- E-commerce platforms
- Airline ticket pricing
- Ride-sharing systems

Key approaches include:

- Rule-based pricing
- Demand forecasting
- Reinforcement learning

Recent research focuses on combining ML + RL for adaptive pricing.

4. Problem Statement

Design a system that:

- Predicts optimal product pricing
- Adapts to demand fluctuations
- Maximizes revenue dynamically

5. Methodology

5.1 Data Generation

Synthetic dataset with:

- Demand
- Competitor pricing
- Seasonal factors

5.2 Feature Engineering

- Lag features (time-series dependency)
- Demand trends

5.3 Machine Learning Models

- Random Forest
- Boost

5.4 Reinforcement Learning

- Q-learning algorithm
- Discrete pricing actions
- Reward-based optimization

6. System Architecture

Modules:

1. Data Collection
2. Data Processing
3. ML Model Training
4. Pricing Engine
5. RL Optimization Layer
6. Visualization

7. Implementation

Tools & Technologies

- Python
- Pandas, NumPy
- Scikit-learn
- Boost
- Matplotlib

Workflow

1. Generate dataset
2. Create lag features
3. Train ML models
4. Evaluate predictions
5. Apply Q-learning
6. Generate optimal pricing

8. Results and Analysis

Model Performance

- Random Forest MAE: (your output)
- XGBoost MAE: (your output)

Observations

- XGBoost performs better in most cases
- Lag features improve accuracy
- RL provides adaptive pricing strategy

Visualization

Include your graph:

- Actual vs Predicted prices

9. Advantages

- Real-time adaptability
- Scalable solution
- Combines prediction + decision-making

10. Limitations

- Synthetic dataset
- Simplified RL model
- Limited real-world constraints

11. Future Scope

- Use real-world datasets
- Implement Deep Reinforcement Learning
- Deploy as web API
- Add user personalization

12. Conclusion

The project successfully demonstrates a hybrid pricing engine using machine learning and reinforcement learning. It shows how intelligent systems can optimize pricing strategies and improve revenue.

13. References

- Scikit-learn documentation
- XGBoost documentation
- Research papers on dynamic pricing
- Kaggle datasets